

Solution Requirements

Date	04 July 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR):** The system allows user to enter socio-economic data, process it, and predict the Human Development Index (HDI). It displays the prediction results through charts and reports for analysis and decision making.
- **Non-Functional Requirements (NFR):** The system should be secure, reliable, user-friendly, and provide fast prediction results. It should ensure data privacy, high performance, and compatability across different devices and web browsers.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users can securely access the application to use the HDI prediction system.	High
2	Authorization levels	The System allows authorized users to enter data, predict HDI, and view results.	High
3	External interfaces	The system provides a user-friendly web interface for entering socio-economic data, displaying HDI predictions, and visualizing charts and reports.	High

4	Transactions processing	The system processes user input, validates the data, performs preprocessing, and generates HDI predictions accurately.	High
5	Reporting	The system generates prediction reports, dashboards, and graphical visualizations of HDI results for analysis.	Medium
6	Business rules, etc.	The system accepts only valid socio-economic data, applies machine learning algorithms for prediction, and ensures accurate processing based on predefined rules.	High
7	Compliance to laws or regulations	The system protects user data, follows data privacy guidelines, and uses datasets only for educational and research purposes.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric/ Acceptance Criteria
1.	Performance & Speed	The system should process user data and generate HDI predictions quickly with minimal delay.	Prediction generated within 2-5 seconds.
2.	Scalability	The system should efficiently handle an increasing number of users and larger datasets without affecting performance.	Supports at least 1000+ records and 50+ concurrent users without significant slowdown.
3.	Security & Data Privacy	Protect user data using secure authentication and access control.	Unauthorized users cannot access data

4.	Reliability & Availability	The system should provide accurate HDI predictions and remain available with minimal downtime.	99% system availability with consistent predict results.
5.	Usability & Accessibility	The application should have a simple, user-friendly interface and be accessible on different devices and browsers.	Users can easily perform predictions without training; supports Chrome; Edge, and Firefox.